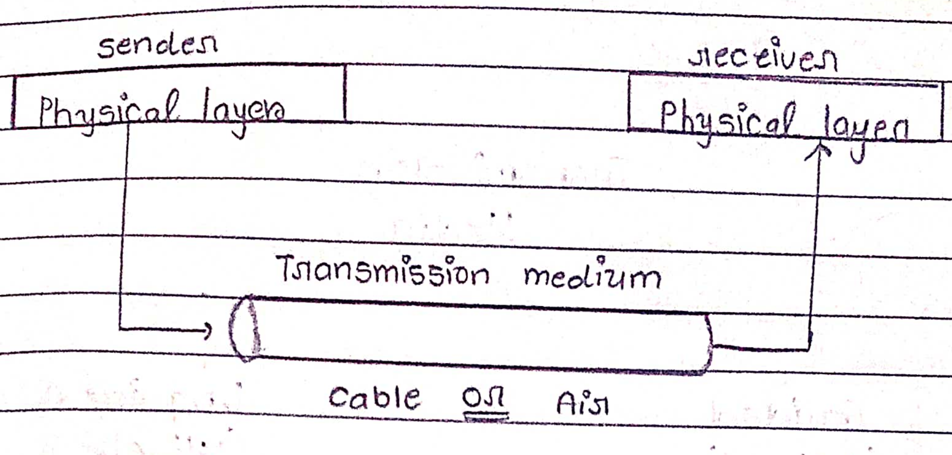
UNIT-3 Transmission Media & Network devies

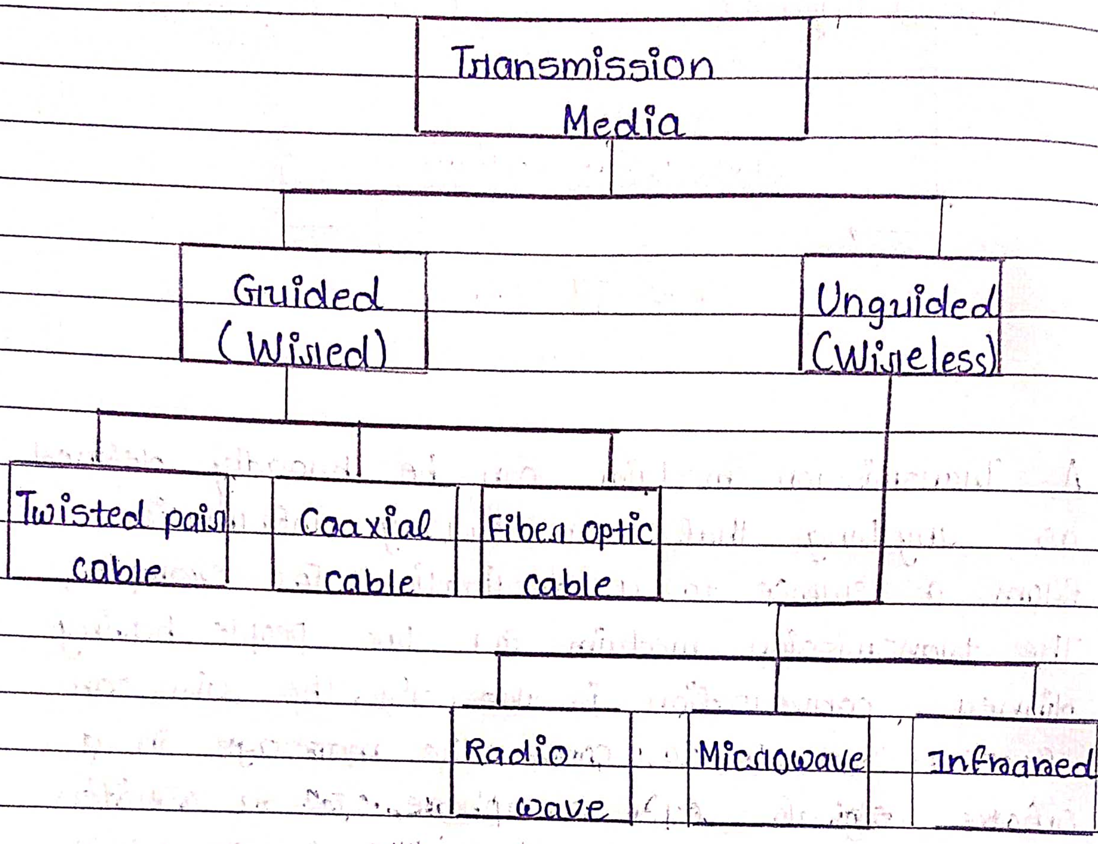
1* Transmission Medium :-



→ A Transmission medium can be broadly defined as anything that can carray information from a source to a destination. For example, The transmission medium for two people having dinner conversation is the air. The air can also be used to conve the massage in a smoke signal or semaphore. For a writen message, the transmission medium might be mail carrier, a truck or a airplane.

→ In data communications the transmission medium is usually a free space, metalic cable or Fiber-optic cable. The information is usually a signal that is the result of a conversion of data from another form.

* 2 Classes of transmission media :



2.1 * Guided Media :

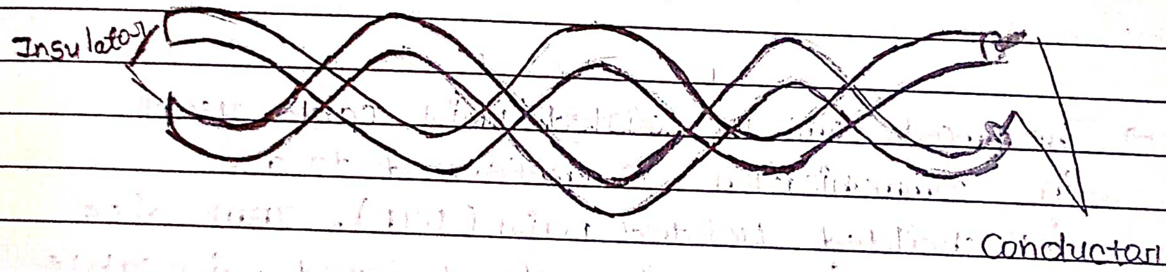
→ Guided Media, which are those that provide a conduit from one device to another, Include twisted-pair cable, coaxial cable, fiber-optic cable. A signal traveling along any of these media is directed and contained by the physical limits of the meduim.

→ Twisted pair and coaxial cable use metalic conductors that accept and transport signal in the form of electric current. Optical fiber is a glass or plastic cable that accepts and transport signal in the form of light.

2.2 * Unguided Media :

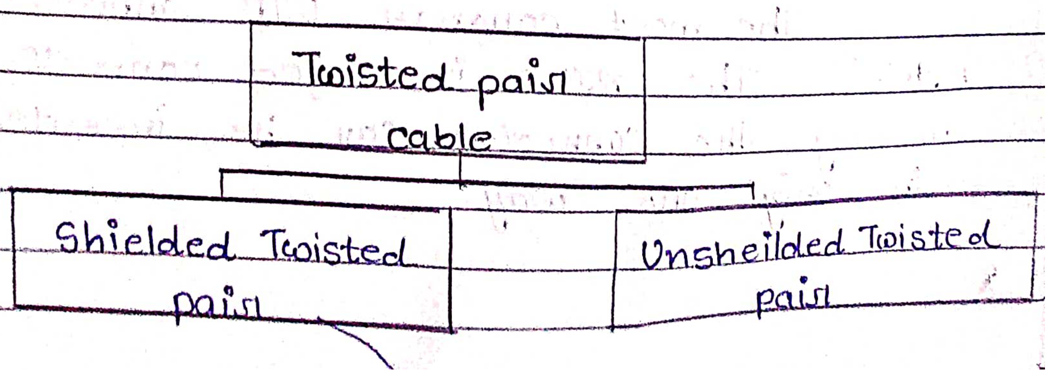
→ Unguided media or wireless comunication, transport electromagnetic waves without using physical conductor, Insted signals are brodcast through air and thus are available to anyone who has a device capable of receiving them.

*3 Twisted -pair cable :

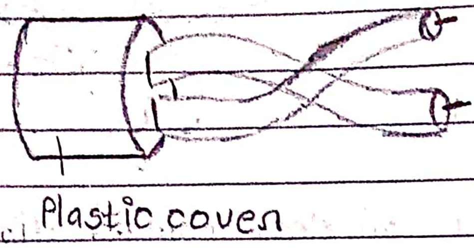


→ It consist of two conductors, each with its own plastic Insulator, twisted together. One of the wire is used to carry signals to receiver, and the other is used only as a ground reference. The receiver uses the difference between the two.

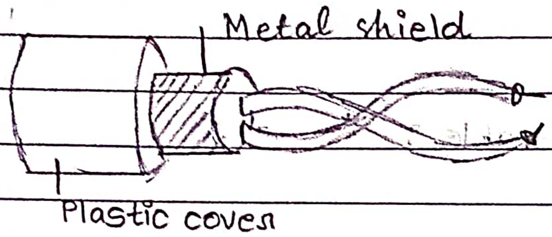
*3.1 Types of Twisted pair cable :



Unshielded Twisted pair :



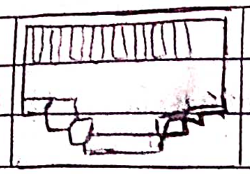
Shielded Twisted pair :



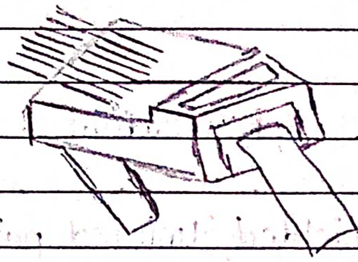
→ The most comman twisted-pair cable used in communication is referred to as "Unsheilded twisted pair (UTP). IBM also produced a version of twisted pair cable for its use, called sheilded twisted pair (STP). STP cable has a metal foil or a braidedmesh covering that encases each pair of Insulator conductor.

*3.2 Connectors :

The most comman UTP connector is "RJ45". The RJ45 is Keyed connector meaning the connector can be inserted in only one way.



RJ-45 Female



RJ-45 Male

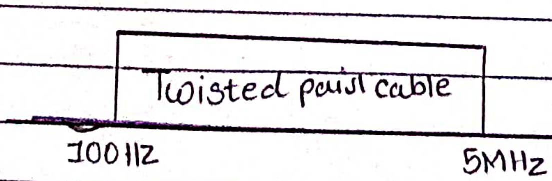
*3.3 Applications :

(•) Twisted pair cable are used in telephone lines to provide voice and data channels.

(•) The local-loop = The line that connects subscriber to the central telephone office commanly consist of unsheilded twisted pair cable.

(•) The DSL lines that are used by the telephone companies to provide high data rate connections also use the high-bandwidth capabality of unsheilded pair cable.

3.4* Frequency Range :

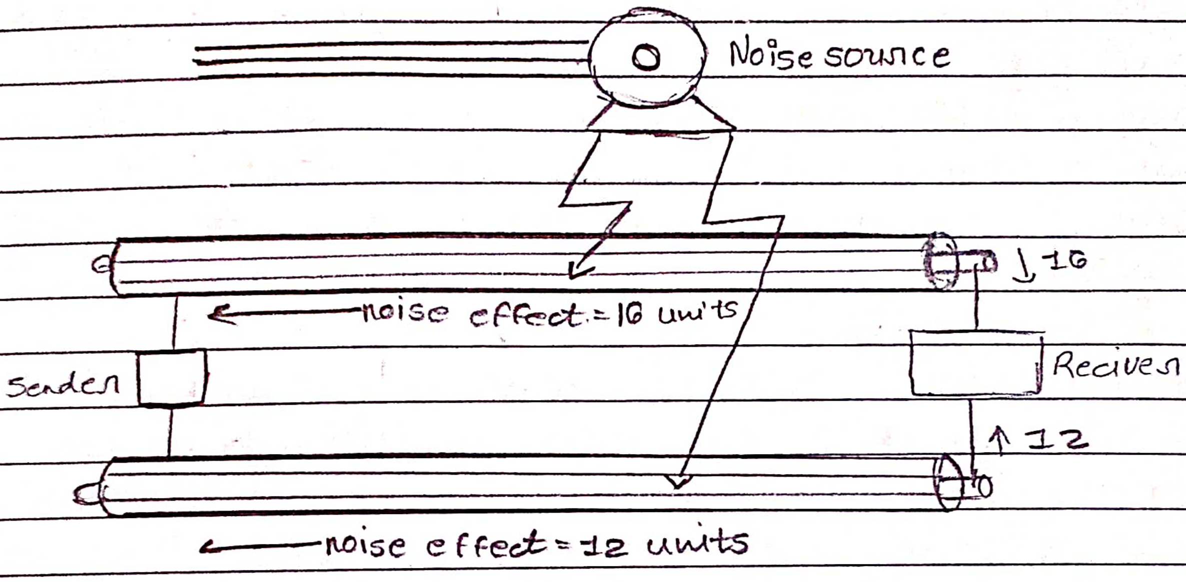


3.5 * Categorys of twisted pair cable :

Category	Specification	Data Rate	USE
1	Unshielded twisted pair used in telephone	<0.1	Telephone
2	Unsheilded twisted pair is Originally used in T lines	2	T-1 lines
3	Improved CAT2 used in LAN's	10	LANs
4	Improved CAT3 used in Token Ring network.	20	LANs
5.	Cable wire is normally AWG 24 with a jacket and outside sheath	100	LANs
5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromegnatic interface.	125	LANs

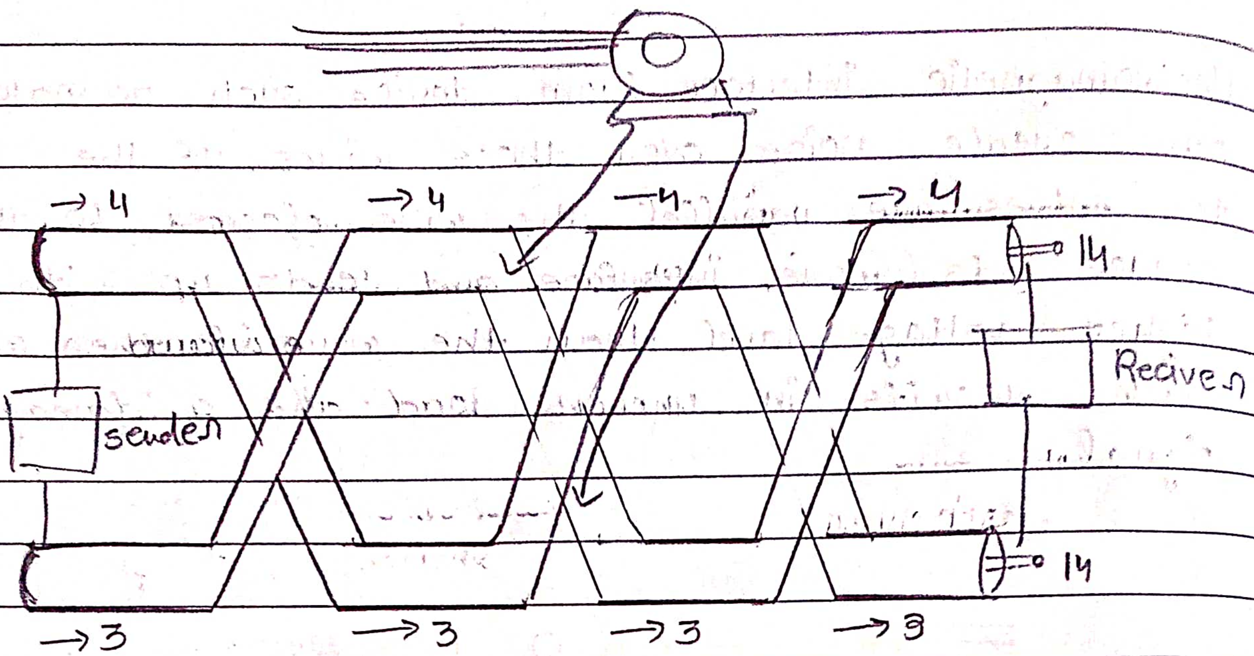
* Why twisted pair cable ?:

-> Electromagnetic interface from device such as motor can create noise over those wires. If the two wires are parallel, the wire closest to the source gets more interface and ends up with higher voltage lavel than the wire further way, which results in uneven load and a damaged signal.



-> Now, If the both wires are twisted than the cumulative effect of the interface is equal on both wires.

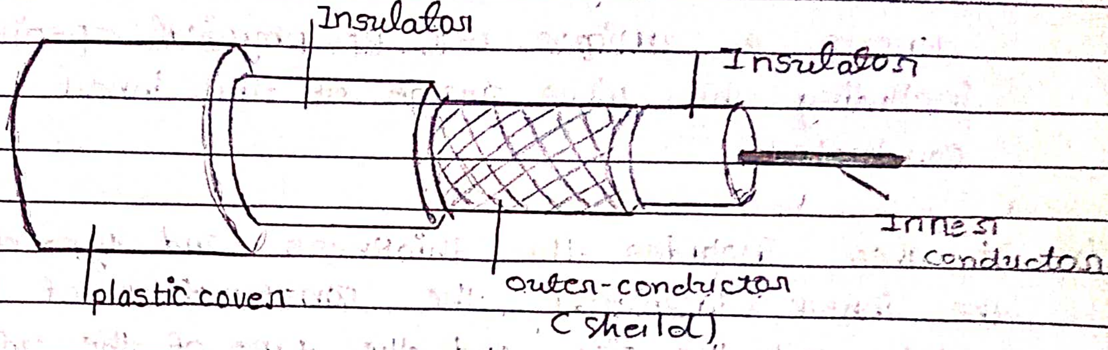
-> The total effect of the noise at reciver is therefor 0. Twisting does not always eliminate the impact of noise, but it does significantly reduce it.



so, $14 - 14 = 0$

4

* Coaxial Cable :



(•) Coaxial Cable carries signals of higher frequency ranges than those in twisted-pair cable, In part because the two media are constructed quailt differently.

(•) Insted of having two wires, coax has a central core conductor of solid or standard wire unclosed in an insulating sheath which is in turn, encased in an outer conductor of metal foil, or combination of two.

(•) The outer metalic wrapping serves both as a sheild against noise and as the second conductor which completes the circuit.

(•) This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by plastic cover as shown in upper fig.

4.1

↳ Standards of coaxial cable :

(•) Coaxial cable are categorized by their Radio Government ratings. Each RG number denotes a unique set of physical specification including the wire gauge of the inner conductor.

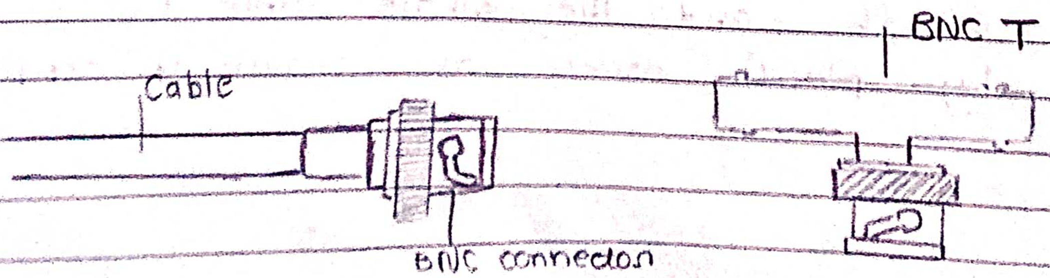
(•) It also includes the thickness and type of the inner insulator, the construction of the sheild and the size and the type of the outer casting.

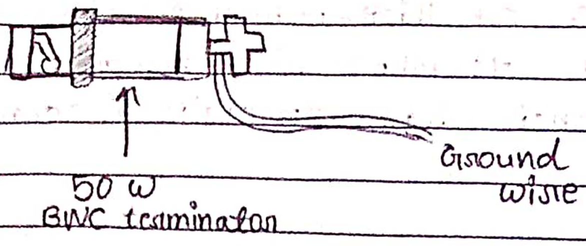
Category	Impedance	Use
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

4.2

* Connectors :

(•) To connect coaxial cable to device, we need connectors. The most comman type of connector used is " Bayonet Neil-concelman (BNC) connector.





(•) BNC connector is used to connect the end of the cable to a device, such as a TV set.

(•) It is also used in Ethernet networks to branch out to a connection to a computer or other device.

4.3
* Applications :

(•) Coaxial cable was widly used in anolog telephone networks where a single coaxial network could carry 10,000 voice signals.

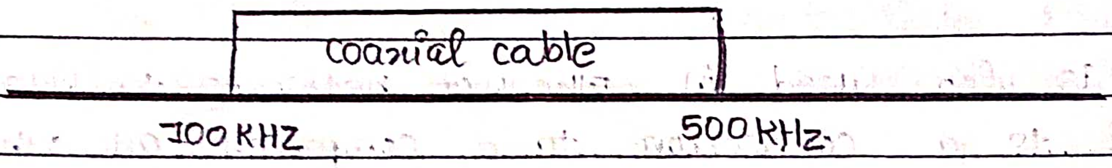
(•) Coaxial cable in telephone networks has largely been replaced today with fiber-optic cable.

(•) Another comman application of coaxial cable is in traditional Ethernet LANs. Becaus of its high bandwidgh, and the consequently high data rate It was chosen for digital transmission in early Ethernet LANs.

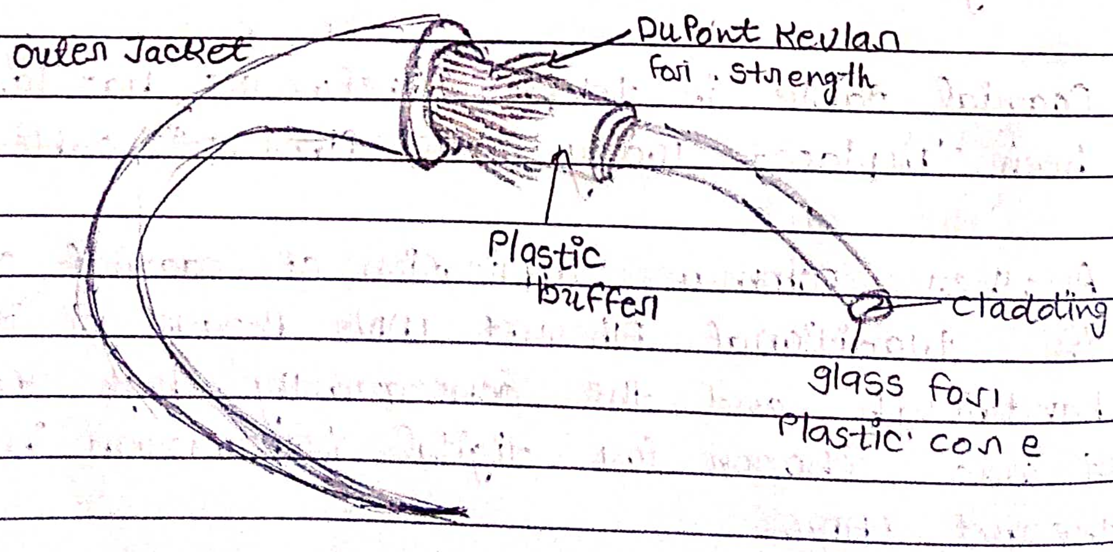
•) The 10base-2 or thin ethernet uses RG-58 coaxial cable with BNC connector to transmit data at 10 Mbps with a range of 185 m.

(•) The 10Base5 or thick ethernet uses RG11 to transmit 10 mbps with range of 5000m. Thick ethernet has specialized connectors.

4.4 Frequency range :



* Optical Fiber cable :



(2) Reflection :

(1) When the angle of incidence becomes greater than the critical angle, a new phenomenon occurres called Reflection.

(3) Critical Angle :

(1) The change in the incident angle results in a refracted angle of 90 degress, with the refracted beem now lying along the horizontal.

(2) The incident angle at this point is known as the critical angle.

* Propagation Modes :

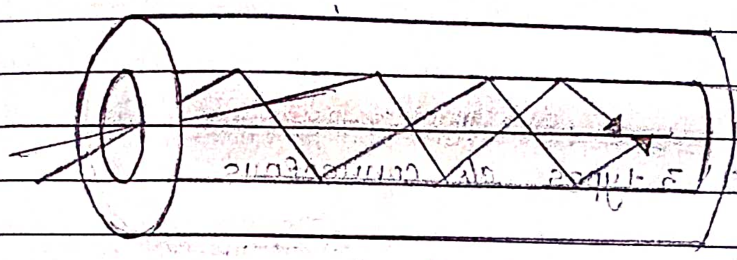
- Mode
  - Multi mode
    - Step-index
    - Graded index
  - Single mode

(1) Multimode :

→ It is named multimode because multiple beams from a light source move through the core in different paths.

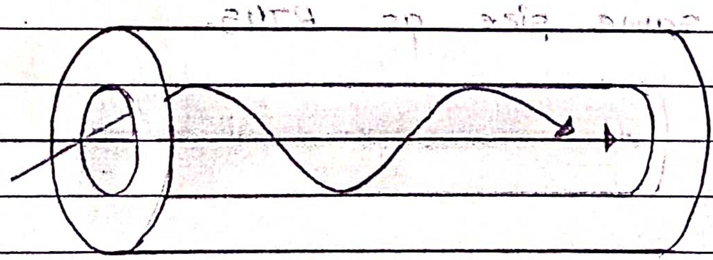
(1) Step index fiber :

(•) A beam of light moves through this constant density in a straight lines until it reaches the interface of core and the cladding.



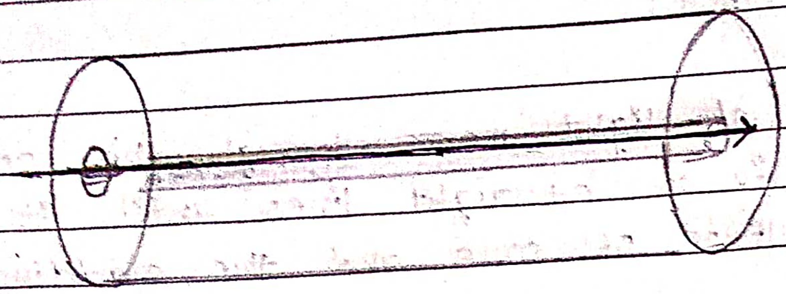
(2) Graded index Fiber:

(•) Graded index fiber is one with varying densities. Density is highest at the center of the core and decreases gradually to its lowest at the edge.



(3) Single Mode :

(•) It uses step index fiber and highly focused Source of light that limites beam to small range of angles, all close to horizental.



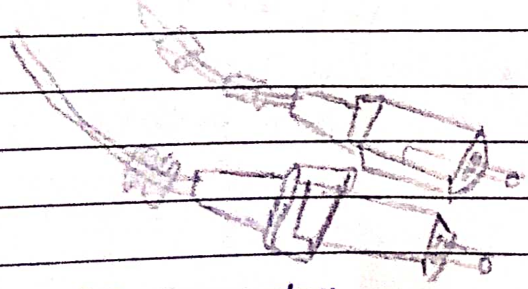
* Connectors:

(•) There are 3 types of connectors.

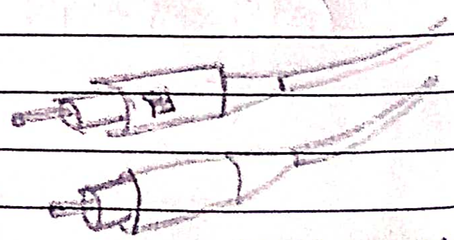
(1) Subscriber channel (SC)
(•) used for cable TV and uses PUSH/PULL locking system.

(2) Straight tip (ST)
(•) Used to connect networking device and uses bayonet locking system.

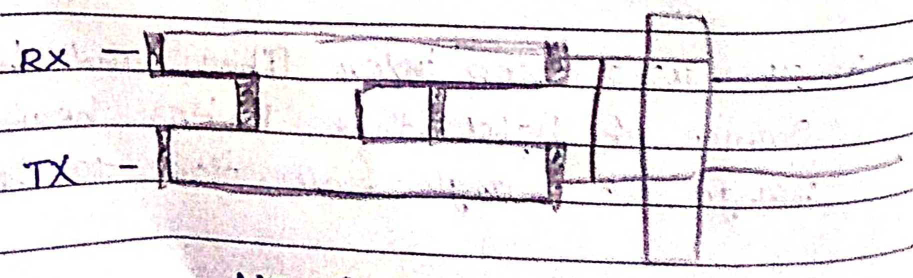
(3) MT-RJ
(•) It is same size as RJ45.



SC connector



ST connector



MT-RJ connector

* Applications :

(•) It is often found in backbone network because its wide bandwidth is cost-effective.

(•) some cable TV companies use a combination of optical fiber and coaxial cable thus creating a hybrid network.

(•) This cost-effective configration since the narrow bandwidth requirement at the user end does not justify the use of optical fiber.

* Advantages :

(•) Higher bandwidth

(•) Less signal attenuation

(•) Immunity to electromagnetic interference

(•) Resistance to corrosive materials.

(•) Light weight

(•) Greater immunity to tapping :

(•) Installation and maintenance.
(•) Unidirectional light propagation.
(•) Cost

* Unguided Media :

(•) This transport electromagnetic waves without using a physical conductor. This type of communication is often refered to as "wireless communication".

(•) Signals are normaly broadcast through free space and thus are available to anyone who has a device capable of receving them.

(1) Radio wave
(2) Micro wave
(3) Infraded wave

400THz

Power voice	Radio Communication Radio, satelite, micro	Infraded light	Ultra Violet light	x gamma, cosmic rays

0 3KHz 300GHz

Visible light (430-750 THz)

* Propagation methods :

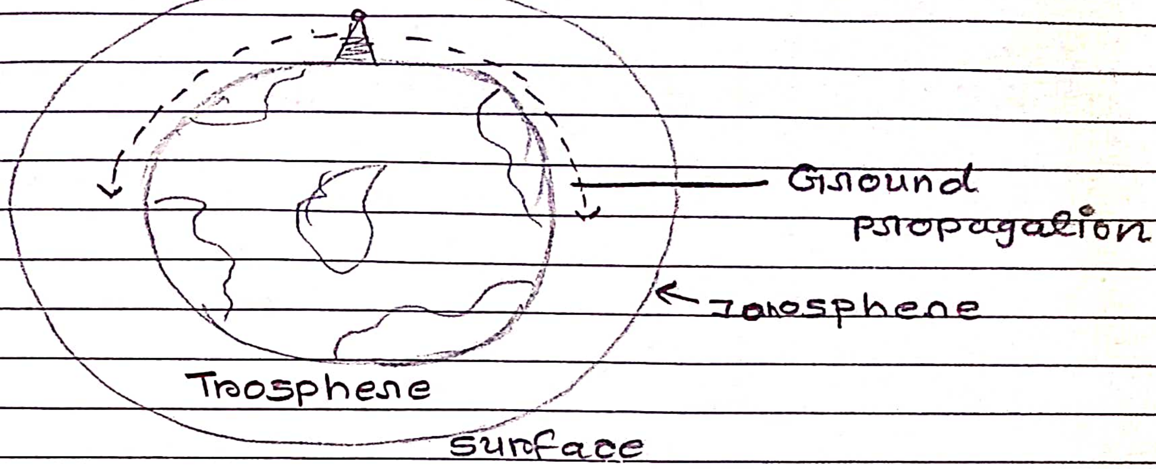
(1) Surface propagation / ground

(2) Tropospheric propagation / sky

(3) Line-of-sight propagation

(4) Space propagation

(1) Surface propagation :/ Ground propagation:



(•) In this, Radio waves travel throuh the lowest portion of the atmosphere, hugging the earth.

(•) At the lowest frequensis, signals emanate in all direction from the transmitting antenna and follow the curvature of the planet.

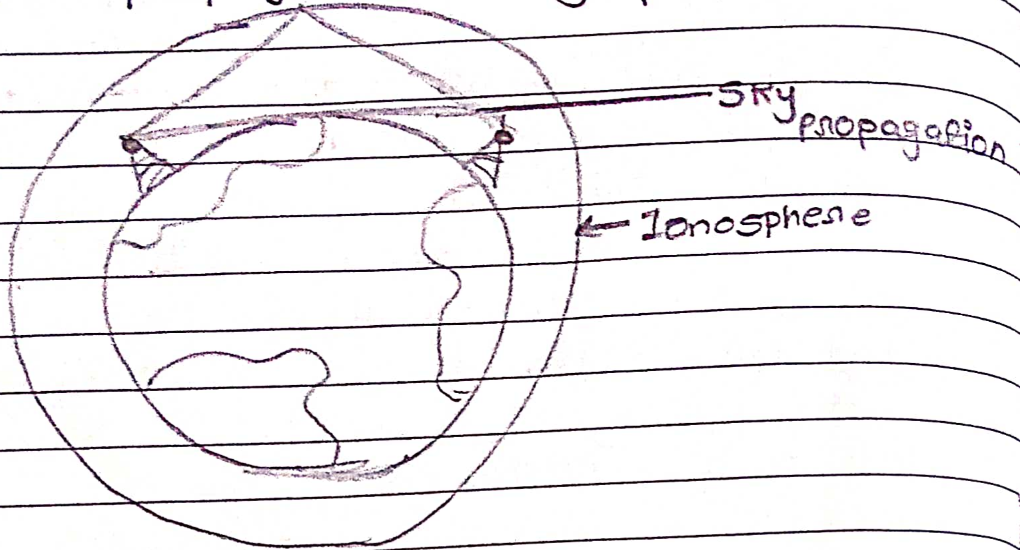
(•) Distance dipantance on the amount of power in the signal : greater the power greater the distance.

→ Application

→ Long range radio navigation

→ Radio beacons and navigational locators.

(2) Troposphenic propagation :/ sky propogation



(•) This works in 2 ways: Either a signal can be directed in a strait line from antenna to antenna. or it brodcast at an angle into the upper layers of the tropospheric where it is reflacted back down to the earth's surface.

(•) The first method is used / requirs the placement of receiver and transmitter be within line of strait distance

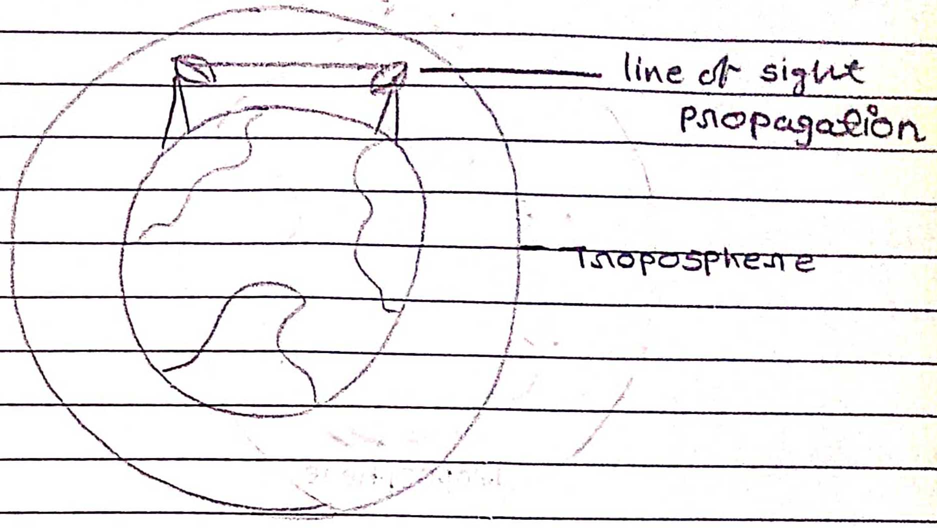
(•) The second method allows greater distance to be coverd.

* Applications:

(•) AM Radio

(•) Citizen band, ship / air craft

* Line-of-sight propagation :



(•) In this, very ~~hi~~ high frequency signals are transmitted in stairt line directly from antenna to antenna.

(•) lion of sight propagation is tricky because radio transmission can not be completly focused. waves emanate upward and downwords as well as forward and can reflect off the surface of the earth.

(•) Reflacted waves that arrive at the receiving antenna later than the direct portion of the transmission can corrupt the recivied signals.

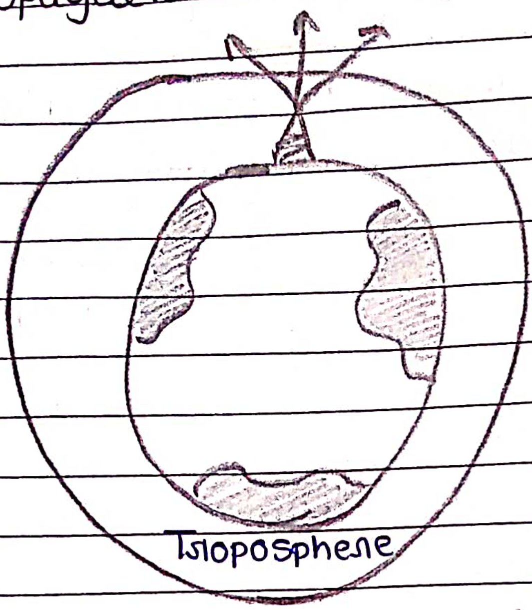
→ Application :

(•) UHF TV, cellular phones, satelite

(•) Radar, satellite

(•) paging, FM radio.

(4) Space propagation :



(i) It utilizes satellite relays in place of atmospheric refraction. A brodcast signal is by an orbiting satellite, which rebrodcasts the signal to the intended received by ~~an orbiting satellite~~ earth.

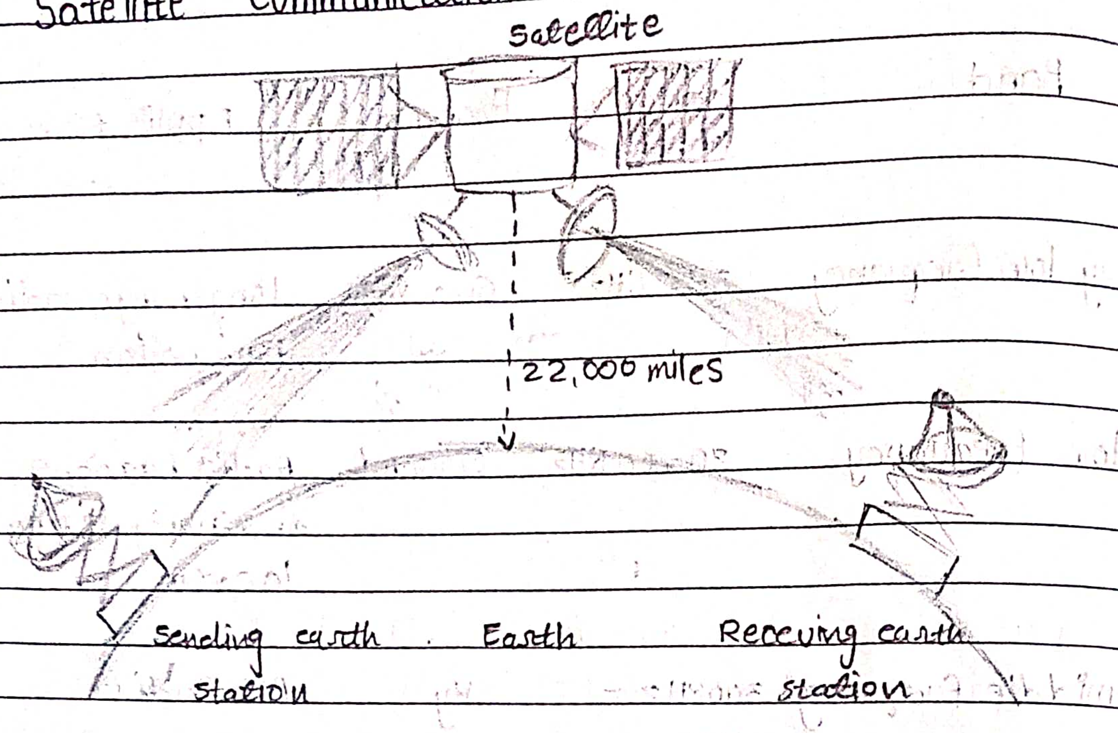
(ii) Which is the distance of the satellite from the earth makes it the equivalent of a super high gain antenna and dramatically increases the distance coverable by a signal.

- Application :

- ~~Distv~~ DishTV
- ~~satlelite~~ Sattelite

Band	Range	Propagation	Application
very low frequancy	3-30 KHz	Ground	Long-range radio navigation
low frequancy	30-300 KHz	Ground	Radio baacons and navigational locators.
middle frequency	300KHz - 3MHz	sky	AM Radio
high frequancy	3-30MHz	sky	citizen band, ship / aircraft
very high frequency	30-300MHz	syky and lionofsight	VHF, FM radio
ultrahigh frequency	300MHz - 3GHz	Line of - sight	UHF TV, cellular phone, paging
superhigh frequency	3-30 GHz	Line of - sight	satellite
extremely high frequency	30-300GHz	Line of - sight	Radar, satellite

* Satellite Communication :



→ Satellite transmission is much like line of sight microwave transmission in which one of the stations is a satellite orbiting the earth.

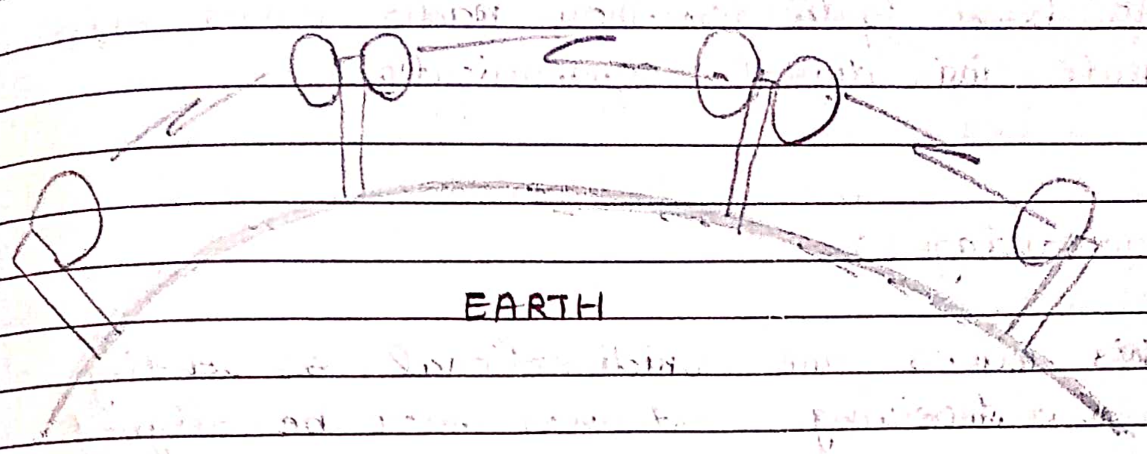
→ In Satellite transmission signals must travel in straight lines.

→ Satellite microwave can provide transmission capability to and from any location on earth no matter how remote.

→ This advantage makes high quality communication available to underdeveloped parts of the world without requiring a huge invesment in ground based infrastructure.

→ Satellite themself are extremely expensive of course but leasing time or frequency on one can be relatively cheap.

* Terrestrial Microwave : (1GHz To 300GHz)



→ Microwaves do not follow the curvature of the earth and there for requires line of sight transmission and reception equipment.

→ Microwave signals propagate in one direction at a time, which means that two frequencies are necessary for two way communication such as telephone conversation.

→ One frequency is reserved for microwave transmission in one direction and the other for transmission in other.

→ Each frequency requires its own transmitter and receiver.

→ Today, both pieces of equipment usually are combained in a single piece of equipment called transceiver.

- Advantages :

→ There is no interface of another pair of aligned antennas.

→ As band is wider than radio waves, high rate for digital communications.

- Disadvantages :

→ This waves are unidirectional, so sending and receiving antennas must be aligned.

→ Repeaters are required for longer distance.

- Applications :

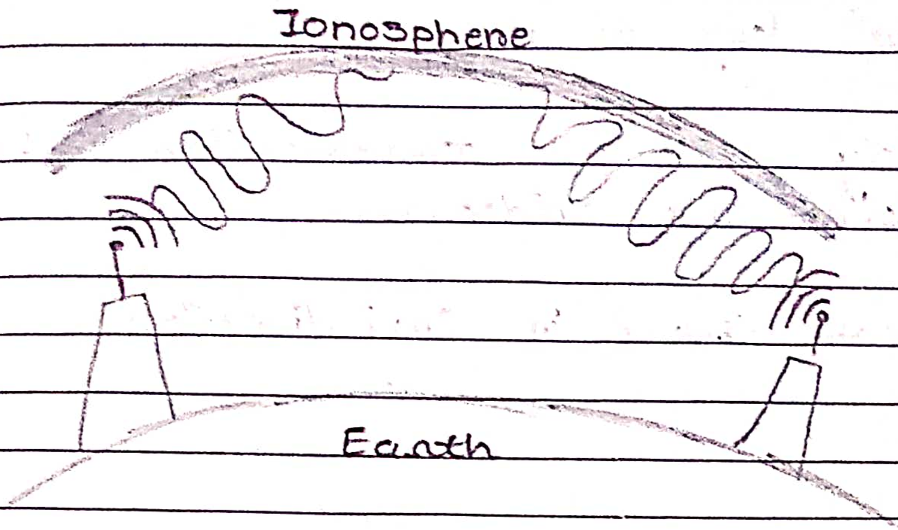
→ Used for unicasting.

→ cellular phones

→ Satellite networks

→ Wireless LANs

* Radio wave transmission :



→ Radio waves are electromagnetic waves occurring on radio frequency portion of the electromagnetic spectrum.

→ These electromagnetic waves are ranging in frequency between 3KHz and 1 GHz

→ A common use is to transport information through the atmosphere or outer space without wires.

→ Radio waves are omni directional means they can travel in all direction from the source. So transmitter and receiver do not need to aligned physically.

→ Radio waves are of "sky propagation."

→ They are easy to generate, can travel long distance and can penetrate buildings.

- Advantages:

→ Travel long distance

→ Widely used for communication in both indore and outdoors.

→ It can travel in Omni direction

- ~~Distan~~
- Disadvantages:

→ Due to the narrow side bands, there is low data rate for digital communication.

→ They can penetrate walls so we can not isolate whether the communication to just inside or outside the building.

- Applications:

→ AM and FM radio

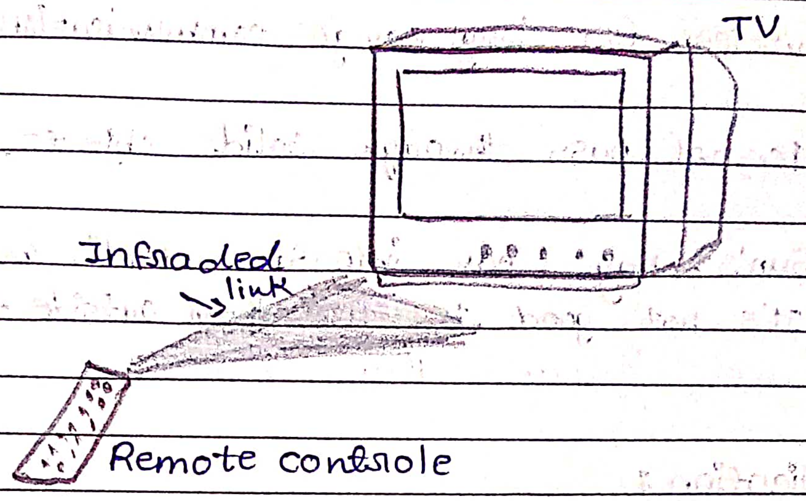
→ GPS service

→ cordless Phone

→ police radio

→ Television broadcast.

* Infraded waves transmission :



(•) These electromagnetic waves are ranging in Frequency between 300 GHz and 400 THz.

(•) They are used for short range communication.

(•) Example : remote control used in televisions,
VCRS
stereos.

(•) They are line of sight propogation.

* Advantages :

→ They are relatively directional, cheap and easy to built.

→ Having high frequency so because of that they are not able to penetrate the walls.

* Disadvantages:

→ Useless for long range communication.

→ Do not pass through solid objects.

→ Sun's ray can interfare radio waves,
It's not good to use them outside the buiding.

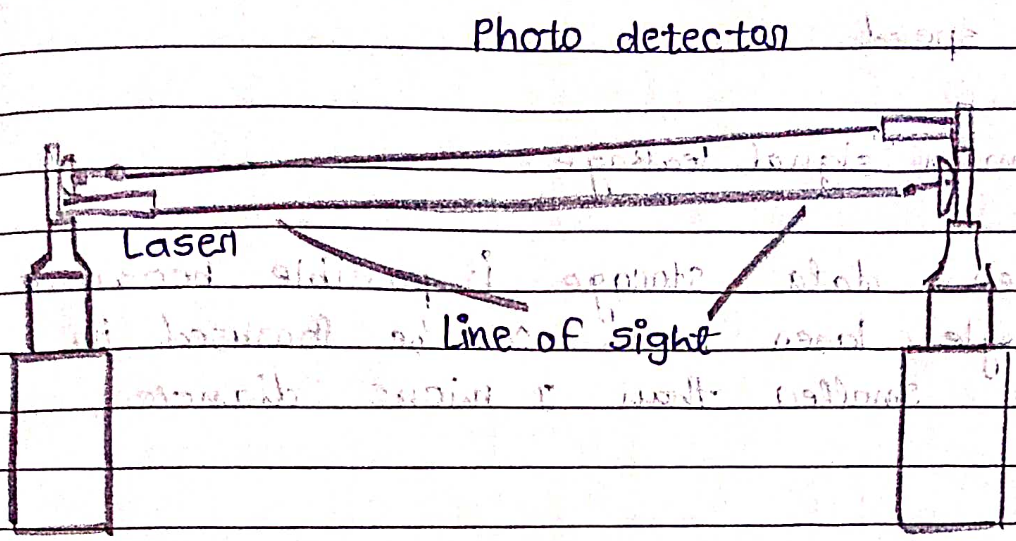
* Application:

→ Remote control of TV, VCR.

→ Thermal efficiency analysis, remote tempretine sensing.

→ military application: target acquisition, survcilance tracking.

* Laser transmission:



(•) Full form: Light Amplification Stimulated Emission of Radiation.

(•) They are undirectional, Therefor this type of system use "line of sight" propogation.

(•) It can Transmit signal to long term.

(•) They can not penetrate obstacles such as walls rain, thick fog. laser beam is distarted by wind, atmosphere tempretune.

(•) A photo detector and laser is set on both sender and receiver side.

* Advantages:

(•) It is safe for data transmission as it is 7mm wide laser which is difficult to tap.

- High bandwidth at low coste.
- High speed.
- minimum signal leakage.
- Higher data storage is possible because single laser beam can be focused in area smaller than 1 micro diameter.

* Disadvantages :

- Maintenance is costly.
- harmful to human beings and often burns thems during contacts.

* Application :

- It has been utilized for mass communication including telephone conversation and even television channels.
- Used in digital data transmission.
- Effectively used for shorted distance communication.

UNIT : 3 Transmission Media & Network Devices

* Network Devices :

→ Definition :

Network devices are essential components in computer networks that facilite communication between different hardware.

→ They can be categorized into two main types :
i > End-devices and ii > Intermediate devices.

→ Types of Network Devices :

- End Devices :
These devices originate or terminate data flow in a network. Example include ; Computers, Printers and Servers.

- Intermediate Devices :
These devices connect end devices and manage data transmission across the network.

Types Listed Below :
i > Repeater
ii > Bridge
iii > Hub
iv > Switch
v > Router
vi > Gateways
vii > Access point

i) Repeaters :

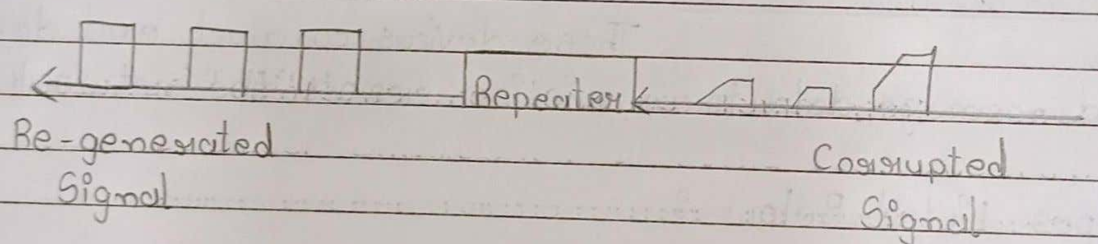
- Repeater is an electronic device which operates only in physical layer.

- Repeaters, installed on a link receives the signals before it becomes too weak or corrupted, it regenerates the original bit pattern and puts the new refreshed copy of signal back on the link.

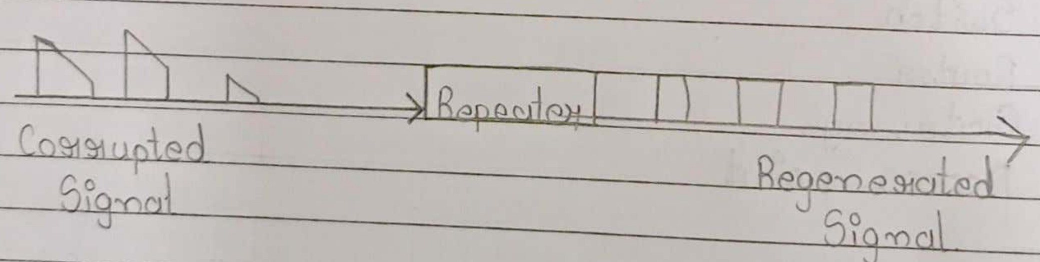
- Repeaters allows us to extend only the physical length of a network.

- It does not change the functionality of the network.

- Figure :



a. Right-to-left transmission



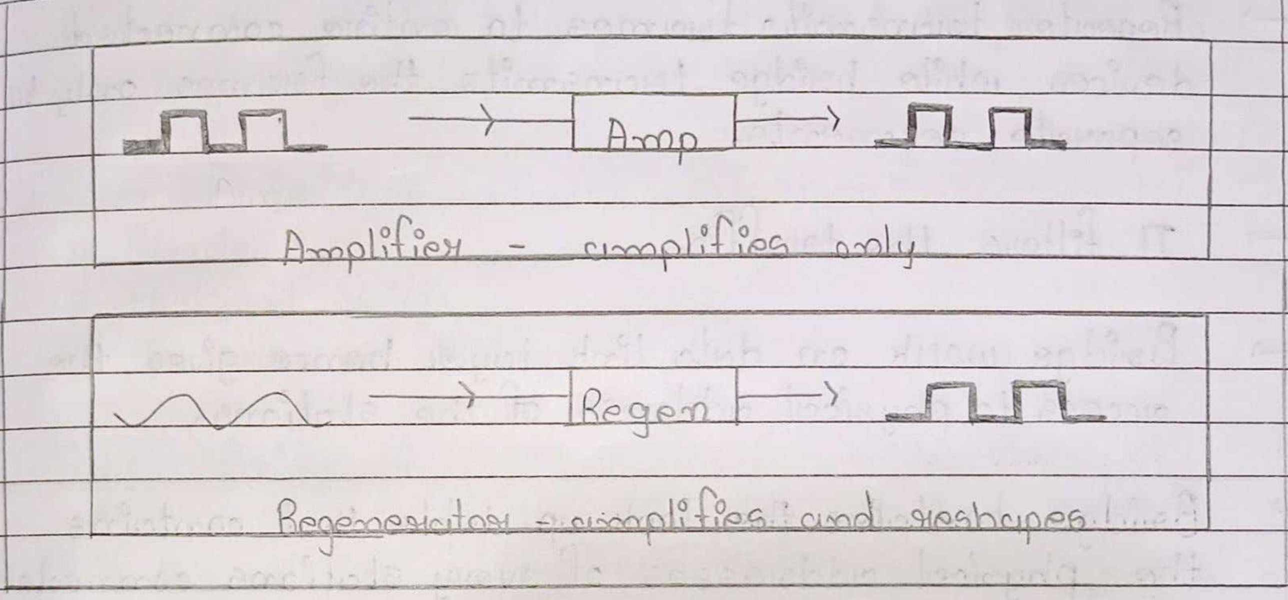
b. left-to-right transmission

- Repeater is not an amplifier.

→ An amplifier can not differentiate the intended signal and noise signal, hence it amplifies equally everything [given as input] into it.

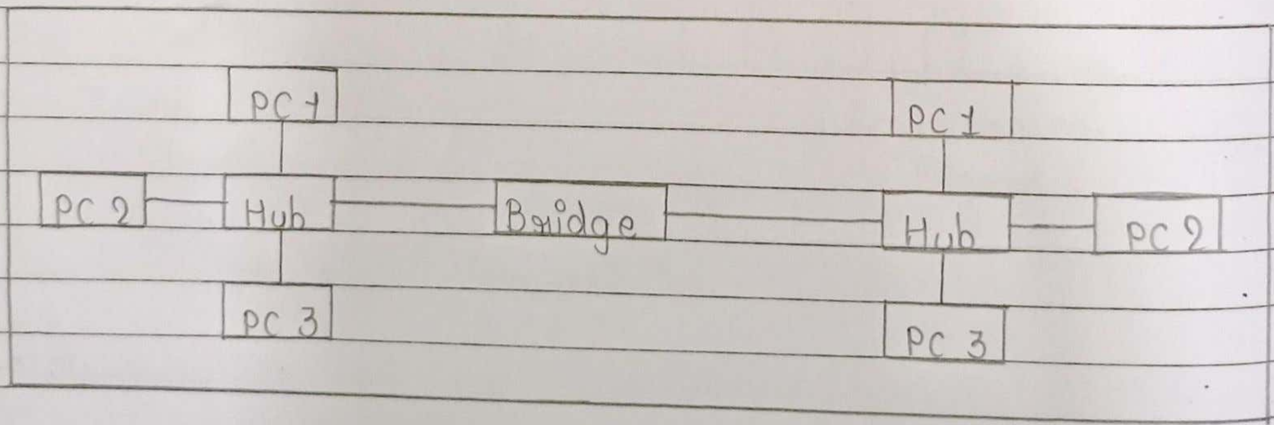
→ While repeater does not amplify the signal instead it regenerates it.

→ Repeaters placed so that a signal reaches it before the noise changes the meaning of any of its bits.



ii> Bridge :

- It operates on both physical and data link layer of OSI model.
- It connects two or more LANs.
- Bridge has a software which keeps the separate traffic for each segment.
- Repeater transmits frames to entire connected devices while bridge transmits the frames only to separate segments.
- It filters the traffic.
- Bridge work on data link layer hence gives the access to physical address of the stations.
- Bridge contains the look up table that contains the physical addresses of every stations connected to it.
- Bridge is also worked for collision resolution.
- Figure :



- In the working of a bridge → when a frame enters to the bridge, it regenerates the signal and it checks the address of the destination and forwards the new copy only to segment to which the address belongs.

- As a bridge found the frame, it reads the address contained in the frame and compares the address with a table of all the stations on both the segments.

- When bridge finds the correct match, it finds to which segment the station belongs and send the frame to only that segment.

- Types of Bridges :
  1. Simple Bridge
  2. Multiport Bridge
  3. Transparent Bridge

1. Simple Bridge :

- It is the most primitive and least expensive types of a bridge.

- It links two segments.

- It contains a table that lists the physical address of all the stations connected with it. Physical address must entered manually.

- In this bridge, updating of device is time consuming when new device is add / removed, the table must be modified at this time.

- So installation and maintenance for this bridge is time consuming and tedious.

2. Multiport Bridge :

↳ It is used to connect more than two LANs.

↳ In this bridge, different tables are created each one holding the physical address of stations reachable through the corresponding port.

3. Transparent Bridge :

↳ It builds the table of physical address on it's own as it performs its bridge function.

↳ Table is automatically built as frames are moved in the networks.

↳ Initially when bridge is installed, it is empty.

↳ As transmission is done the table is updated accordingly.

↳ The first packet transmitted by each station, the bridge makes entries inside the table with corresponding segment. So at last the table is completed with all details.

↳ The bridge has "Self updating features".

iii) Hub :

↳ Hub is used to create connections between stations in physical star topology.

↳ Large number of computers can be connected in single or multiple LANs.

↳ Hub is a central network device that connects network nodes. So is also referred as concentrator.

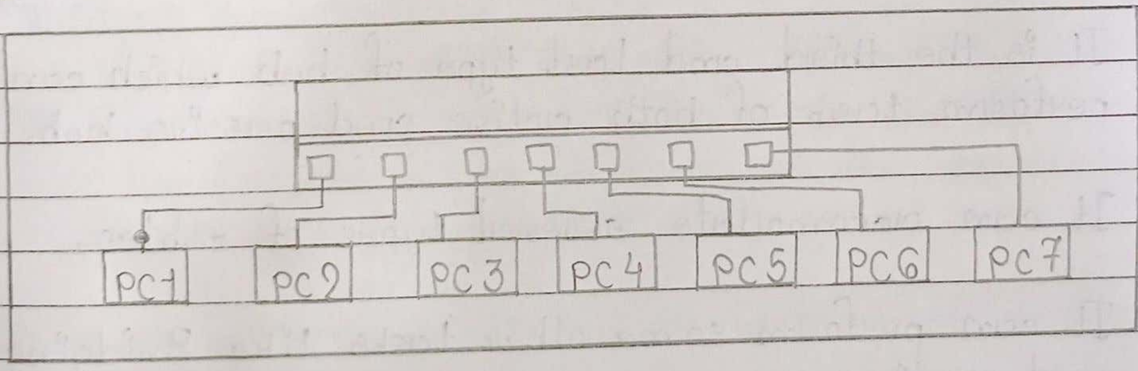
↳ Hub enables central network management.

↳ It provides connections for several different media types like : coaxial, fiber optic, twisted pair.

↳ Hubs are available in 8 / 16 / 24 ports.

↳ It provides high speed communication.

↳ Figure :



↳ Types of Hub :

1. Passive Hub
2. Active Hub
3. Intelligent Hub

1. Passive Hub :

↳ It provides the physical connections between the attached devices.

↳ It do not amplify or regenerates the signals passes through hub.

↳ It do not require electricity to run.

2. Active Hub :

↳ It acts like repeaters, it regenerates and retransmits the signals.

↳ Because of multiple ports are available in it, it also called Multiport Repeaters.

↳ It need electric power to run.

3. Intelligent Hub :

↳ It is the third and last type of hub which can perform task of both active and passive hub.

↳ It can accomodate several types of cables.

↳ It can perform some other tasks like Bridging and routing.

↳ It increases the speed and effectiveness of total network thus makes the performance of whole network fast and efficient.

↳ It is intelligent because capable of doing hub management and switching functionality.

iv) Switch :

↳ Switch provide the bridging functionality with greater efficency.

↳ It acts as a multiport bridge to connect different devices or segments in a LAN.

↳ It operates in data link layer of OSI model [also called layer 2 switch].

↳ Switch has buffer for each link connected with it.

↳ When it receives the frame, it stores the frame in the buffer of receiving link and checks address to find outgoing link.

↳ If outgoing link is free the switch sends the frame to that particular link.

↳ Two different characteristics of switch :

i) Store and forward switch → it stores the frame in input buffer until the whole frame is arrived.

ii) Cut through switch → It forwards the frame to the output buffer as soon as the destination address is received.

↳ Difference Between Layer 2 switch & Layer 3 switch.

Layer-2 Switch	Layer-3 Switch
↳ Forwards data based on MAC address.	↳ Routes data based on IP addresses & forwards data based on MAC addresses.
↳ Cannot route data between VLANs.	↳ Can route data - between VLANs.
↳ Requires little to know configuration.	↳ Requires a more complex configuration.
↳ Relatively in expensive.	↳ More expensive.
↳ Very fast.	↳ Slower.

v> Router :

↳ A router is hardware device designed to receive, analyze and move incoming packets to another network.

↳ It operates in physical, data link and network layer of the OSI model. But most active in network layer.

↳ Routers are able to access network layer address [IP address] of the device.

↳ Simple function of router is to receive the packet from one connected network and pass them to a second connected network.

↳ Routers also perform the traffic directing functions on the Internet.

↳ A packet sent from a station on one network to a device on a neighboring network goes to the jointly device - router which forwards the packet to the destination network.

↳ Routers consult with routing table when packet is ready to be forwarded.

↳ Characteristics of Router :

- Least - cost routing :
  Router finds the shortest path for the packet which is fastest, cheapest, reliable and secure.

↳

- Non-adaptive routing :
  In which, once a pathway to a destination has been selected, the router sends all packets for that destination along that one route.

- Adaptive routing :
  In this, Router may select a new route for each packet. Router sends the packet depending on which route is most efficient at that moment.

* Types of Router :

i) Static Router :

- In it, routing table information are entered manually.
- In case of change in the connection, it cannot update automatically.
- It is more secure, It always uses the same route.

ii) Dynamic Router :

- In it, routing table is created and updated automatically.
- When there is a change in the connection, routing table is updates using any of the routing protocols.

vi) Gateways :

↳ Gateways operates in all seven layer of OSI model.

↳ A gateway is a hardware device that acts as a "gate" between two networks.

↳ It is also called as protocol convertor.

↳ It is generally used to connect two different network systems.

↳ Routers transfers packets only across networks using similar protocols while a gateway can accept a packet formatted for one protocol and converts in to a packet formatted for another protocol before forwarding it.

↳ It may be a router, firewall, server, or other device that enables traffic to flow in and out of the network.

vii) Access Point :

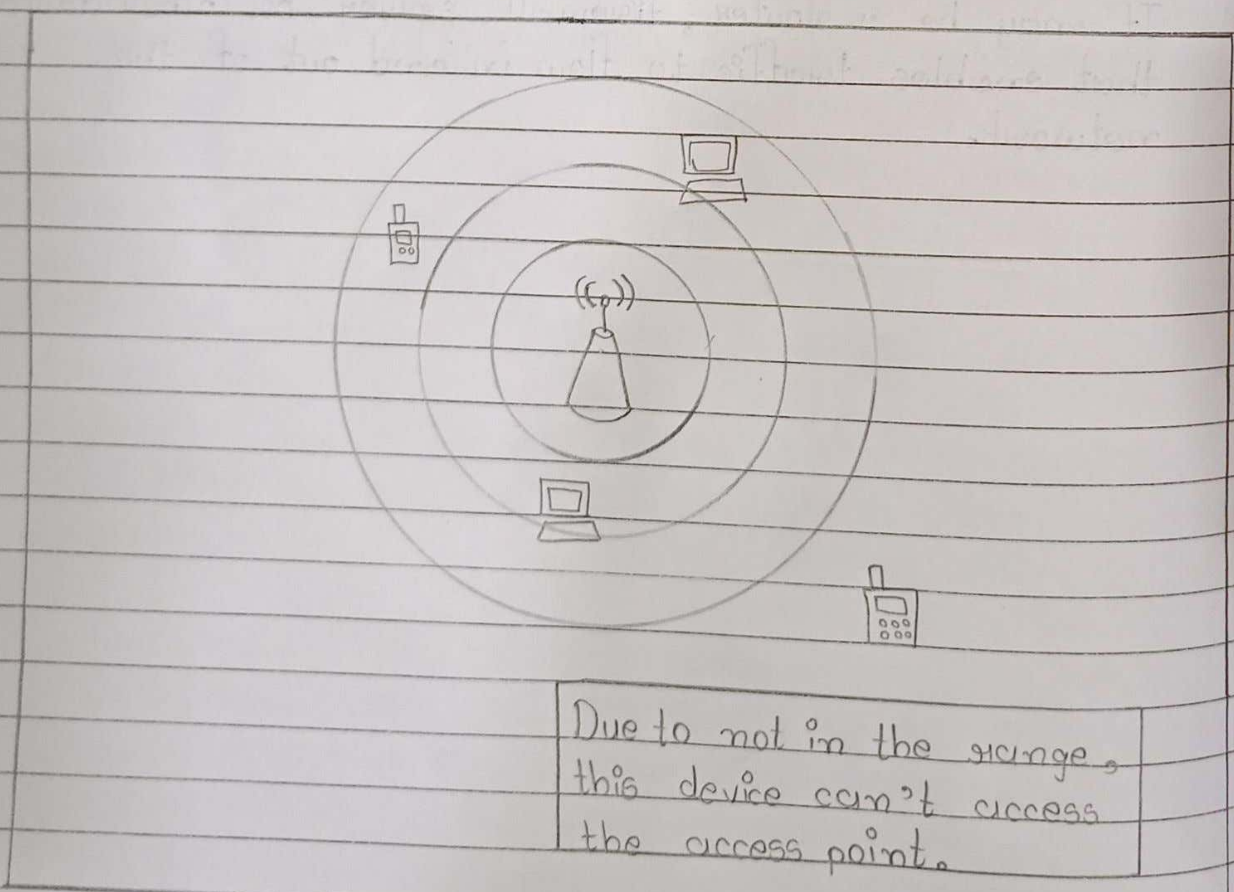
↳ An access point is a device that allows wireless devices to connect to a wired network using Wi-Fi.

↳ Most access points have built-in routers, while others must be connected to a router in order to provide network access.

↳ Access points can be found in many places, including houses, businesses, and public locations.

↳ Every access point has its own range. Any device placed within the range of access point can only use the services of that.

↳ Figure :



↳ Working :

- Wireless access point performs the technique called 'modulation' to transfer the data receives in or out.

- It takes the modulated radio frequency voltages highs and lows transmitted from a nearby wireless network card, and turns them into 1s and 0s instead.

- These 1s and 0s assembled into packets and forwarded to the destination.